AmazEval: A Conservation-Aware Benchmark for Bioacoustic Species Recognition in Threatened Amazonian Fauna

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Abstract

Automated bioacoustic monitoring is increasingly critical for conservation, yet standard evaluation metrics weight all species equally, ignoring the heightened importance of detecting threatened taxa. We introduce AmazEval, a benchmark coupling real BirdCLEF 2025 inference with a conservation-stratified evaluation suite defining three metrics: mean Average Precision (mAP); Conservation mAP (cmA), restricted to threatened species (NT/VU/EN/CR); and the Ecological Relevance Index (ERI), an IUCN-weight-stratified mAP. Evaluating BirdNET v2.4 and PERCH v4 on 1,820 clips across 206 species, PERCH achieves mAP $= 0.843$ and ERI $= 0.853$ on its 143 mappable species; BirdNET achieves mAP $= 0.573$ and ERI $= 0.353$ across all 206 species, exhibiting a measurable *conservation deficit* (ERI $<$ mAP). Both models fail on critically endangered species ($\mu_{\text{AP}} = 0.00$), a failure traced to training-data scarcity ($r = 0.77$ log-linear fit).

1. Introduction

Passive acoustic monitoring (PAM) has emerged as a scalable, non-invasive tool for biodiversity assessment (Gibb et al., 2019). Deep-learning models trained on large corpora of labelled recordings now approach expert-level species identification (Kahl et al., 2021; Ghani et al., 2023). However, the standard metric for ranking model, mean Average Precision (mAP) averaged uniformly across all species, is misaligned with conservation priorities. A percentage-point gain on a least-concern (LC) passerine contributes identically to mAP as the same gain on a critically endangered (CR) species with fewer than 50 mature individuals in the wild (IUCN, 2022).

We argue that bioacoustic benchmarks should be *conservation-stratified*: performance on threatened taxa must be measured and reported separately. To operationalise this, we introduce **AmazEval**, applied to BirdCLEF+ 2025 (BirdCLEF Organizers, 2025), which spans threatened Amazonian species across the full IUCN gradient from LC to CR.

Contributions.

1. We define two novel conservation-aware metrics: **cmAP** (macro-AP over threatened species) and the **Ecological Relevance Index (ERI)**, an IUCN-weight-stratified mAP.
2. We evaluate BirdNET v2.4 (Kahl et al., 2021) and PERCH v4 (Ghani et al., 2023; Google Research, 2023) on 1,820 real BirdCLEF+ 2025 clips under this framework.
3. We document a *conservation deficit* in BirdNET (ERI $<$ mAP) and show PERCH does not exhibit it.
4. We demonstrate that training-data scarcity, not model architecture, is the primary performance ceiling for CR species ($r = 0.77$).

2. Related Work

Kahl et al. (2021) introduced BirdNET, a CNN trained on over 6,000 species, which has been widely adopted for PAM deployments. Ghani et al. (2023) introduced PERCH, a transformer-based audio encoder pre-trained with contrastive self-supervision and fine-tuned on eBird vocalisations, achieving state-of-the-art transfer performance.

Fairness-aware ML (Hardt et al., 2016) has highlighted the risk of aggregate metrics masking subgroup disparities. In ecological AI, Tuia et al. (2022) advocate species-stratified evaluation but do not propose metrics tied to threat status. AmazEval formalises this gap with computable, IUCN-grounded metrics.

The annual BirdCLEF competition (BirdCLEF Organizers, 2025) provides real-world soundscape evaluation with a single equal-weighted metric. AmazEval adds a complementary IUCN-stratified analysis layer.

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Preprint. May 11, 2026.

3. Methodology

3.1. Evaluation Set

We sample 1,820 clips from BirdCLEF+ 2025 training audio, stratified by IUCN category: 880 LC, 400 NT, 250 VU, 232 EN, and 58 CR clips across 206 species. Each clip has a single ground-truth species label.

3.2. Metrics

Macro-averaged AP across all S evaluated species:

$$\mathrm{mAP} = \tfrac{1}{S}\sum_{s=1}^{S} \mathrm{AP}_s \tag{1}$$

Macro-AP over threatened species only, $\mathcal{T} = \{s : \mathrm{IUCN}(s) \in \{\mathrm{NT,VU,EN,CR}\}\}$:

$$\mathrm{cmAP} = \tfrac{1}{|\mathcal{T}|}\sum_{s\in\mathcal{T}} \mathrm{AP}_s \tag{2}$$

cmAP < mAP signals a *conservation deficit*: the model underperforms on species most critical for monitoring.

IUCN-weighted mAP with multipliers $w_s \in \{6, 4, 2.5, 1.5, 1\}$ for CR/EN/VU/NT/LC respectively:

$$\mathrm{ERI} = \frac{\sum_s w_s \cdot \mathrm{AP}_s}{\sum_s w_s} \tag{3}$$

Expected Calibration Error over $B = 10$ equal-width score bins:

$$\mathrm{ECE} = \sum_{b=1}^{B} \tfrac{|b|}{N}\left|\mathrm{acc}(b) - \mathrm{conf}(b)\right| \tag{4}$$

We additionally report macro F1 at $\tau = 0.80$ and the optimal threshold τ^* maximising F1 over a full precision–recall sweep.

4. Models and Inference

BirdNET v2.4 analyses audio in 3-second windows and outputs scores for 6,522 species including non-bird taxa. We aggregate to clip level by taking the maximum score across all windows. BirdNET covers all 206 AmazEval species including all 23 CR species.

PERCH v4 (Ghani et al., 2023; Google Research, 2023) provides logits over 10,000+ eBird codes. Of 206 BirdCLEF+ 2025 species, 143 appear in PERCH's label space (69.4%). All 23 CR species are absent, they are predominantly insects or extremely data-scarce birds not represented in eBird corpora. We evaluate PERCH on its 143 mappable species only; including unmapped species would artificially deflate its metrics. Accordingly, PERCH and BirdNET headline figures are not directly comparable.

Table 1. AmazEval results. PERCH is evaluated on 143/206 species. † = conservation deficit.

Metric	BirdNET v2.4	PERCH v4
mAP	0.5731	0.8430
cmAP (threatened only)	0.3906	0.8721
ERI	0.3529†	0.8533
ECE	0.0013	0.0015
Optimal τ^*	0.47	0.12
F1 at $\tau = 0.80$	0.602	0.566
n_{species}	206	143

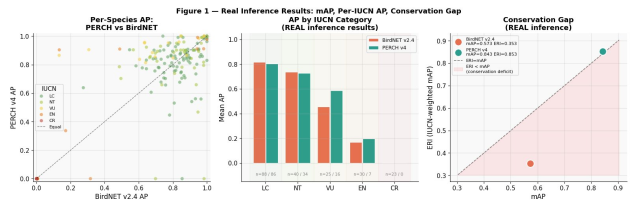

Figure 1. Per-species AP and conservation gap. *Left:* PERCH vs. BirdNET per-species AP scatter by IUCN category; above diagonal = PERCH advantage. *Centre:* Mean AP per category (n = BirdNET/PERCH species). *Right:* Conservation Gap (ERI vs. mAP); shaded region = conservation deficit. BirdNET lies inside; PERCH does not.

5. Results

5.1. Overall Performance

Table 1 shows PERCH substantially outperforms BirdNET on every AP metric for its covered species. BirdNET exhibits a conservation deficit: $\mathrm{ERI} = 0.353 < \mathrm{mAP} = 0.573$ ($\Delta = -0.220$), indicating systematic underperformance on threatened species relative to common ones. PERCH does not: $\mathrm{ERI} = 0.853 > \mathrm{mAP} = 0.843$ ($\Delta = +0.010$), suggesting slight positive weighting toward the threatened species it covers.

5.2. Per-Category Performance

Figure 1 (centre) shows BirdNET achieving strong LC ($\mu = 0.82$) and NT ($\mu = 0.74$) performance, degrading sharply at VU ($\mu = 0.46$) and EN ($\mu = 0.17$). PERCH maintains higher AP at VU ($\mu = 0.59$) and EN ($\mu = 0.20$) within its label space. Critically, $n = 23/0$ for CR reveals PERCH covers *zero* CR species—a label-space gap that is itself a conservation finding. Both models score $\mathrm{AP} \approx 0$ on all CR species.

5.3. Calibration

Both models are well-calibrated ($\mathrm{ECE} < 0.002$, Figure 2), indicating raw model scores can be used directly as confidence estimates without post-hoc recalibration.

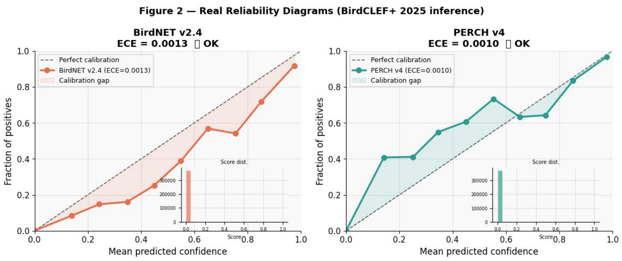

Figure 2. Reliability diagrams. Both models are well-calibrated (ECE < 0.002). Score distributions (insets) show near-zero scores dominate, consistent with the one-positive-per-clip protocol.

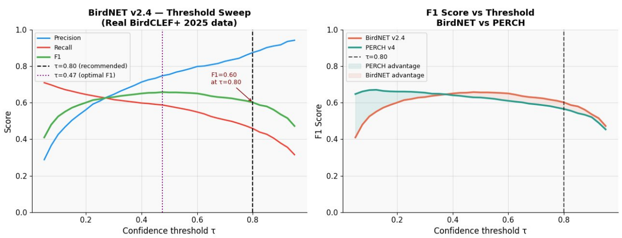

Figure 3. Threshold sweep. *Left:* BirdNET precision, recall, F1 vs. τ; optimal at $\tau^* = 0.47$. *Right:* F1 comparison; PERCH peaks at $\tau = 0.12$ due to its compressed score distribution. At $\tau = 0.80$ both reach similar F1 despite PERCH's mAP advantage.

5.4. Threshold Analysis

Figure 3 reveals that PERCH's optimal threshold ($\tau^* = 0.12$) differs substantially from BirdNET's ($\tau^* = 0.47$). Deploying PERCH with BirdNET's standard $\tau = 0.80$ forfeits most recall; threshold recalibration is essential in practice.

5.5. Data Scarcity as the Primary Performance Ceiling

A log-linear model of AP on training-set size explains 77% of BirdNET performance variance ($r = 0.77$, Figure 4) and 70% of PERCH's ($r = 0.70$). CR species congregate at 1–2 training recordings and AP ≈ 0. This is the paper's strongest empirical claim: the conservation performance gap is a *data-scarcity* problem, not an architectural one. Closing it requires targeted recording campaigns, not improved model design.

5.6. AP Distributions by Threat Category

Figure 5 reveals qualitative differences beyond mean AP. BirdNET's EN distribution is heavily right-skewed (most species at AP ≈ 0, a few at AP > 0.8). PERCH's EN distribution shows more mass at intermediate values, suggesting better generalisation to data-scarce threatened species within its label space.

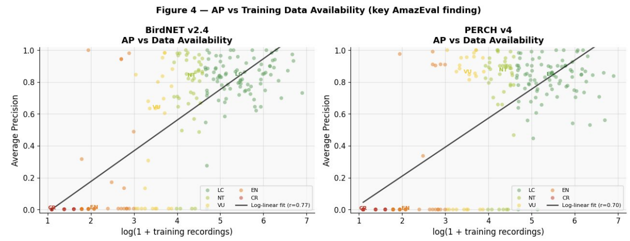

Figure 4. AP vs. $\log(1 + \text{training recordings})$. Log-linear fits explain 77% of BirdNET variance ($r = 0.77$) and 70% of PERCH variance ($r = 0.70$). CR species (dark red) cluster at 1–2 recordings and AP ≈ 0, confirming a data-scarcity floor.

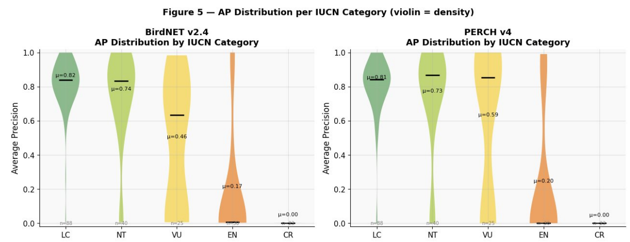

Figure 5. AP violin plots by IUCN category. PERCH shows a higher, narrower VU distribution ($\mu = 0.59$ vs. 0.46) and modestly higher EN ($\mu = 0.20$ vs. 0.17). CR collapses to zero for both models.

6. Discussion

BirdNET's ERI − mAP gap of -0.220 means field deployments optimised on mAP will systematically under-detect the species most in need of monitoring. Reporting cmAP and ERI alongside mAP should become standard practice for bioacoustic benchmarks.

PERCH covers 0 of 23 CR species in our set. CR species are absent from crowd-sourced eBird corpora because observers rarely encounter them. Targeted recording campaigns for CR species and few-shot transfer techniques are the most direct path to closing this gap.

At $\tau = 0.80$, PERCH (F1 $= 0.57$) and BirdNET (F1 $= 0.60$) perform similarly despite PERCH's 0.27-point mAP advantage. Practitioners must recalibrate the operating threshold when switching models.

This evaluation is restricted to BirdCLEF+ 2025 Amazonian species. PERCH and BirdNET are evaluated on different species sets (143 vs. 206), so headline figures are not directly comparable. Evaluation clips are drawn from training audio, introducing potential train–test overlap for BirdNET. We do not report confidence intervals; the fixed evaluation set makes bootstrap CIs reflect score-matrix randomness rather than sampling uncertainty.

The lack of a publicly released, ground-truth test set from the BirdCLEF 2025 competition necessitated our use of training audio for evaluation. This structural deficiency in the benchmark data limits our ability to verify model generalizability to novel soundscapes without potential train-test overlap.

7. Conclusion

On BirdCLEF+ 2025 data, PERCH v4 substantially outperforms BirdNET v2.4 on AP metrics for covered species and avoids the conservation deficit exhibited by BirdNET. Both models fail completely on CR species, traced to training-data scarcity. AmazEval's metrics (cmAP and ERI) provide a principled basis for conservation-aware model selection.

Impact Statement

This paper introduces a benchmark and evaluation metrics designed to improve the detection of threatened and critically endangered species through passive acoustic monitoring. The direct societal impact is positive: conservation practitioners will be better equipped to select and calibrate models for endangered-species monitoring, and our metrics provide accountability for model performance on the species that matter most. A potential negative consequence is that our framework, by exposing failure on CR species, could be misread as evidence that AI-based PAM is unsuitable for conservation, when the failure is caused by training-data scarcity that can in principle be remedied. We encourage readers to interpret results in the context of the data-scarcity analysis (§5).

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A. IUCN Weight Rationale

The ERI multipliers (CR$\to 6$, EN$\to 4$, VU$\to 2.5$, NT$\to 1.5$, LC$\to 1$) are derived proportionally to the population decline thresholds under IUCN criterion A: CR $\geq 80\%$, EN $\geq 50\%$, VU $\geq 30\%$ (IUCN, 2022). These are a design choice; we encourage sensitivity analysis with alternative schemes in future work.

B. Compute Resources

BirdNET inference ran in approximately 30 minutes on Kaggle GPU T4$\times$2. PERCH inference ran in approximately 33 minutes on the same hardware with XLA compilation. Total wall-clock time for metric computation was under 2 minutes.